



CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE
NAME

CLASS

INDEX
NUMBER

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BIOLOGY
Paper 2: Core Paper

8875/02
26th August 2014
2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are two sections in this paper.

Section A]

Answer all questions

Section B]

Answer **ONE** question. Answer each part on a **separate** piece of paper.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

For Examiner's Use	
Section A	40
1 [10]	
2 [12]	
3 [12]	
4 [6]	
Section B	20
5a / 6a [7]	
5b / 6b [7]	
5c / 6c [6]	
TOTAL	60

Section A

Answer **all** questions in this section.

- 1** All cells are surrounded by a cell surface membrane. There are also membrane systems within the cell, as well as several types of membrane-bound organelles. Some of these have a single membrane surrounding them, while others have two.

All cell membranes have a similar basic structure but the proportions of the various molecules present vary between membranes and even between the inner and outer layers of a membrane.

Fig. 1.1 shows a generalised cell surface membrane.

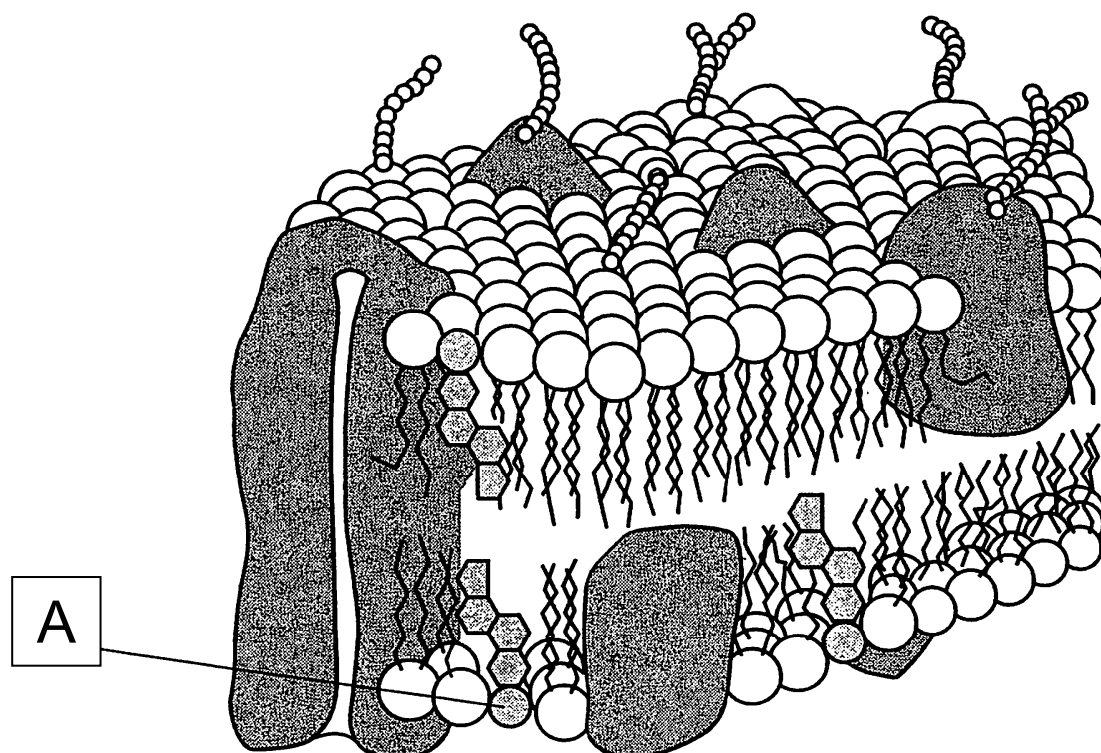


Fig. 1.1

With reference to Fig. 1.1,

- (a) (i)** identify the molecule A and describe its function.

[2]

Molecule A	
Function	

- (ii) With reference to Fig. 1.1, state one way in which the outer surface membrane is different from the inner surface membrane.

.....
[1]

- (iii) state one functional property resulting from this difference

.....
[1]

- (b) Table 1.1 shows the approximate percentage of protein in different membranes.

Table 1.1

Organelle	Membrane	Protein as a percentage of dry weight.
chloroplast	outer membrane	54
chloroplast	inner membrane	70
mitochondrion	outer membrane	55
mitochondrion	inner membrane	80

- (b)(i) using data from Table 1.1, describe one similarity in the distribution of proteins within the two membranes of chloroplast and mitochondrion and suggest a reason.

Similarity	
Reason	

[3]

- (b) (ii) Describe and explain one physical way in which the inner membrane of a mitochondrion is adapted to its function.

.....

[3]

[Total: 10]

2 Fig. 2.1 shows the elongation phase of the cellular process of transcription.

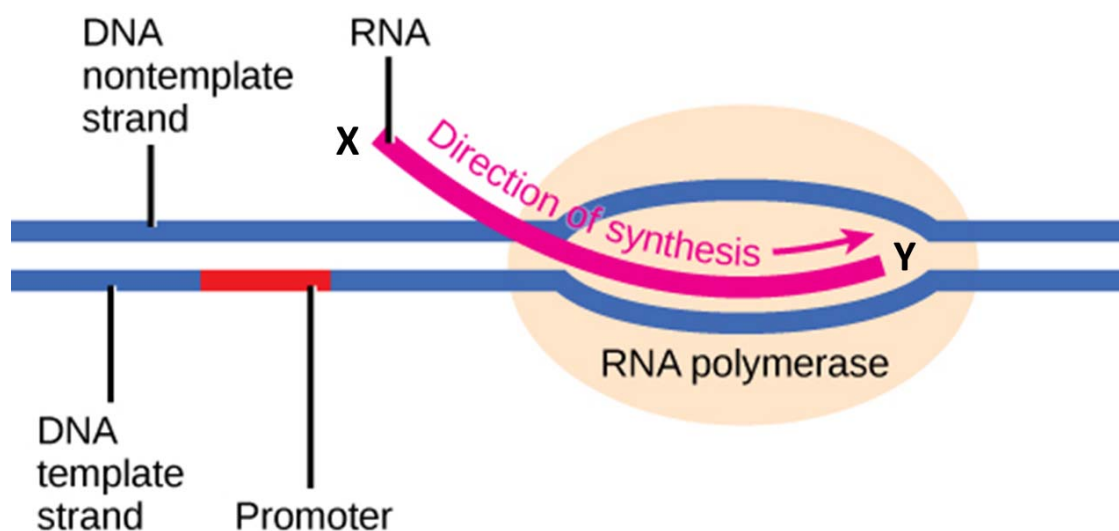


Fig. 2.1

(a)(i) Identify the directionality of the RNA transcript at ends X & Y.

X:

Y:

[1]

(ii) State what are the substrates which are not shown in Fig. 2.1 that RNA polymerase allows in into its active site for the process of elongation to take place.

.....[1]

(iii) The elongation of RNA transcripts requires energy. Explain how this energy is derived.

.....

.....[1]

Protein kinases are a class of enzymes that are involved in the triggering of the cell cycle. Kinases add phosphate to molecules, and the modification can serve as a "switch" to turn events in the cell on or off.

Cyclin dependent kinases (Cdk) regulate the cell cycle. Fig. 2.2 shows a representation of Cdk.

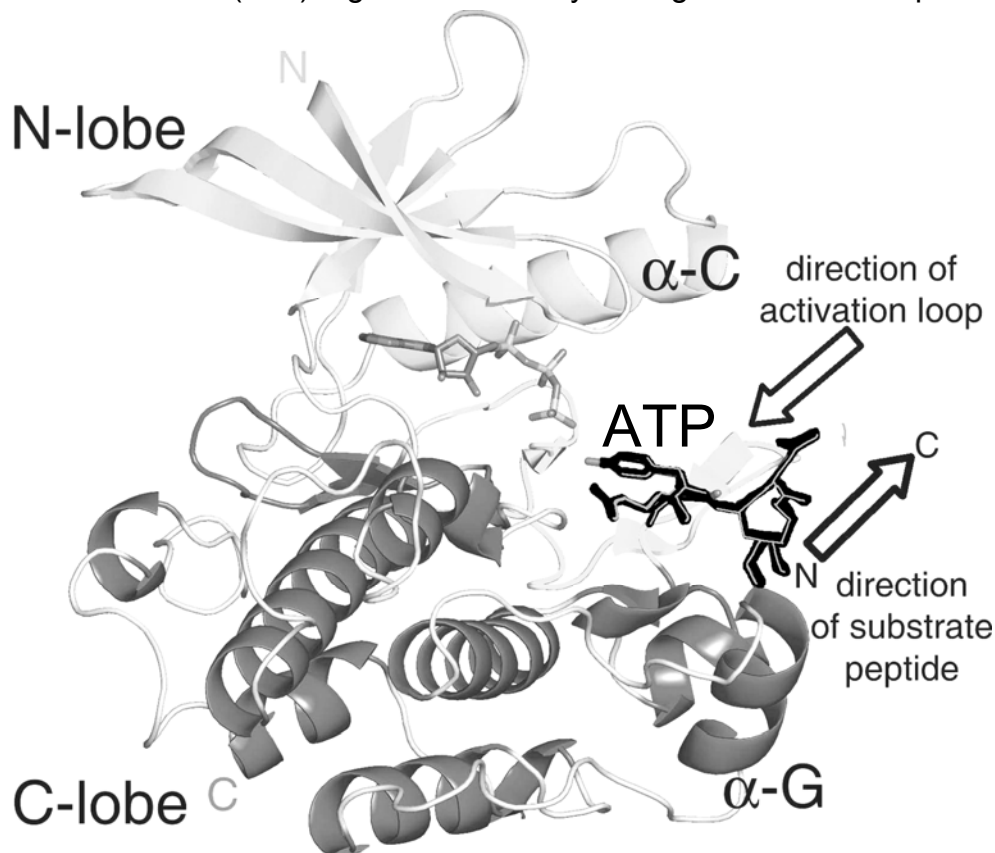


Fig. 2.2 Anatomy of a Cdk catalytic domain. The substrate peptide, docked onto the activation loop, and an ATP analog are shown in black. α -Helix-C, which contains residues critical for catalysis, and α -helix-G, which is involved in docking protein substrates, are labeled.

(b) Describe how specialised enzymes are structurally specific to their substrate

.....

.....

.....[2]

With reference to Fig. 2.2,

(c) explain the specific significance of the following parts of protein kinase

Feature	Significance
N-Lobe and C-Lobe	
α -C and α -G	Catalytic R Groups and binding of the R groups
directions of activation loop and substrate peptide	

Cyclin-dependent kinases (Cdk) are a family of protein kinases first discovered for their role in regulating the cell cycle.

(d) Suggest how Cdk may be regulated in its action

.....
[1]

Cdk works always in conjunction with cyclins (another cofactor in the regulation of the Cell cycle) the complexes that are formed when these two are combined makes Cdk active. Fig. 2.3 shows the relative levels of active Cdk (protein kinase activity) during the cell cycle.

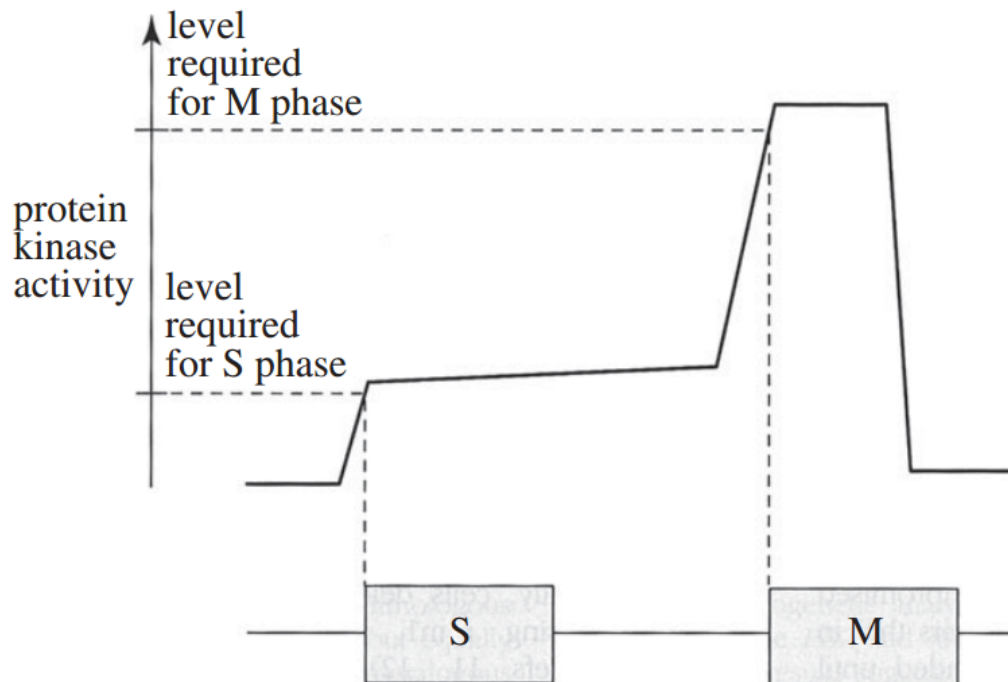


Fig 2.3

(e) Explain what takes place in the gap in-between S phase and M (cell division)

.....

[2]

(f) Explain how CDK is used in the regulation of the cell cycle

.....

[2]

[Total: 12]

3 Fig. 3.1 shows the plasmid pEOD.

Plasmid pEOD is a cloning vector used in creating recombinant DNA. It contains two selectable marker genes, Kan^R (kanamycin resistance gene) and Amp^R (ampicillin resistance gene). A multiple cloning site (MCS) is located within the Kan^R gene.

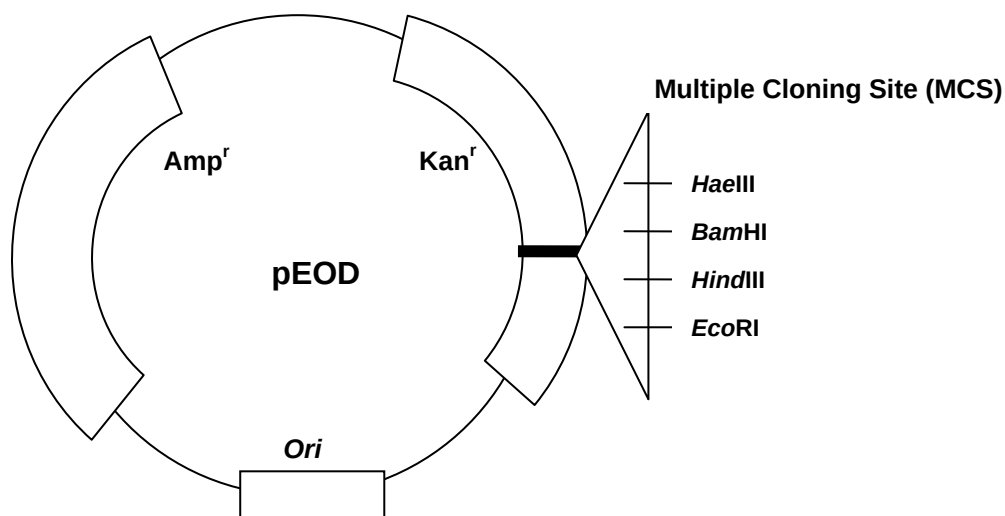


Fig. 3.1

A target sequence of human DNA is replicated using the Polymerase Chain Reaction (PCR).

The target DNA sequence is shown in Fig. 3.2.

5' - GGATCCATCCCGATCGATCC TTTAAGCTAGCTAGGATCC - 3'
3' - CCTAGGTAGGGCTAGCT AGGAAATTCGATCGATCCTAGG - 5'

Fig. 3.2

(a) Design two 8-base long primers for the sequence to be replicated.

.....
.....[2]

(b) State three differences between the process of PCR and natural DNA replication in cells.

.....
.....
.....
.....[3]

After the target DNA has been subjected to PCR, it is cleaved using restriction enzymes and then inserted into pEOD. Fig. 3.3 shows the types of restriction enzymes that can be used to cleave pEOD at the Multiple Cloning Site (MCS) shown in Fig. 3.1

Restriction enzyme Specific target base sequence of DNA

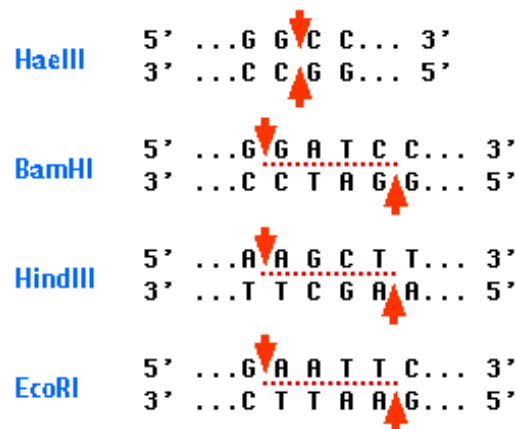


Fig. 3.3

- (c)(i)

State the restriction enzyme used to cleave the target DNA sequence before it is inserted into vector pEOD.

.....[1]
- (ii)

Outline the steps that must subsequently be followed to produce a recombinant plasmid.

.....

.....

.....

.....[3]
- (d)

Describe and explain how bacteria transformed with this recombinant plasmid are identified using selective agar media.

.....

.....

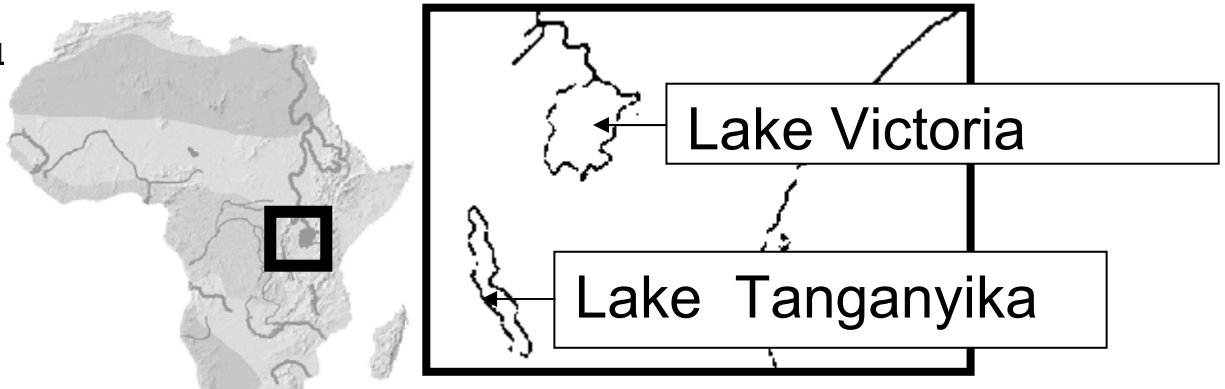
.....

.....[3]

[Total: 12]

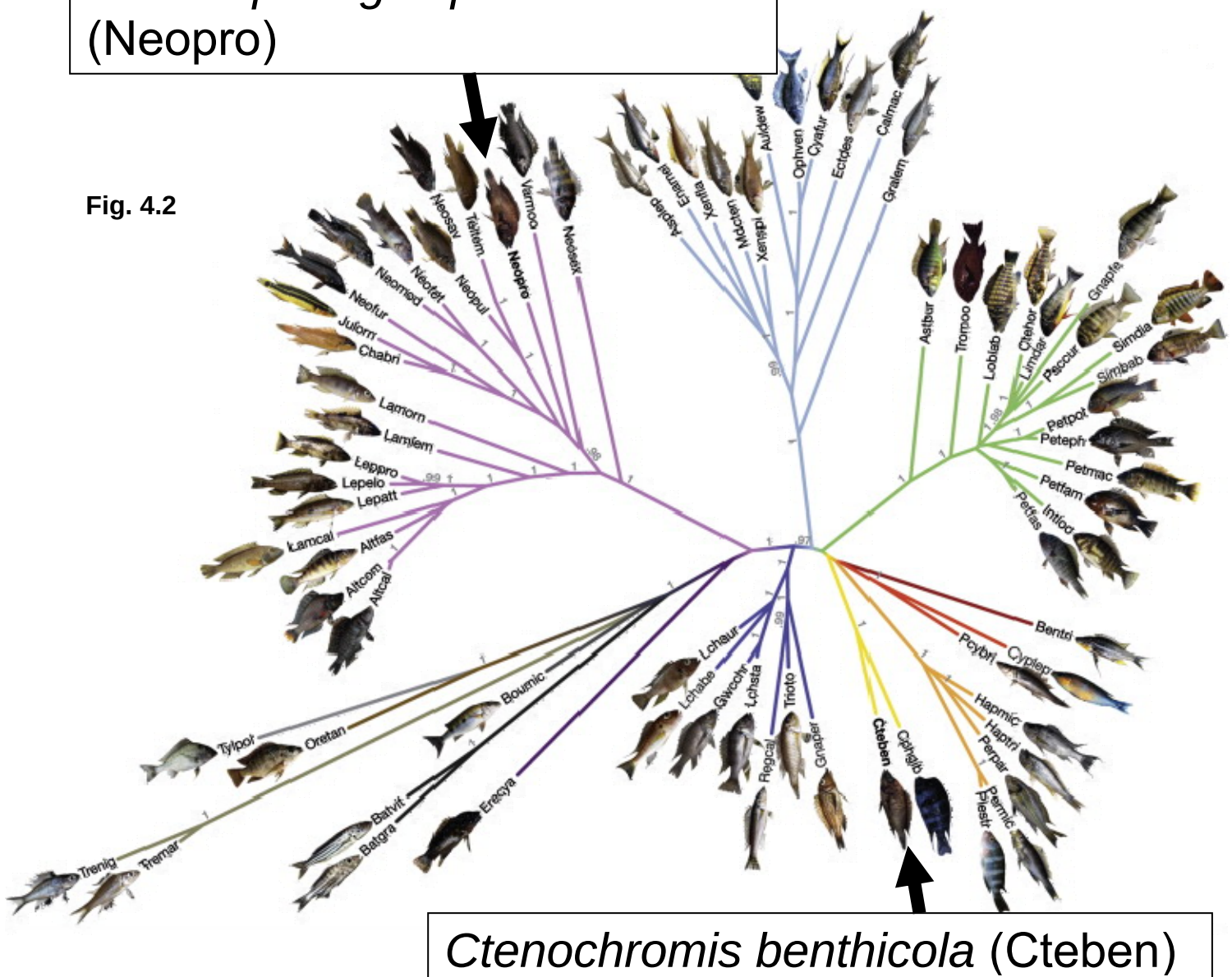
- 4 Cichlids are a fast expanding group of African freshwater fish which is both large and diverse. At least 1,650 species have been scientifically described, making it one of the largest vertebrate families. New species are discovered annually, and many species remain undescribed.

Fig 4.1



Neolamprologus prochilus.
(Neopro)

Fig. 4.2



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Fig 4.1 Map of East Africa showing the Great Lakes. Lake Victoria (LV) and Lake Tanganyika (LT) which is the oldest lake in East Africa

Fig 4.2 The Cichlid Species DNA comparative phylogeny of the 71 cichlid species (**Family: Cichlidae ; Order Perciformes**) of Lake Tanganyika

(a) Explain the possible evolutionary processes that could have led to the many species seen in Fig 4.2.

.....
.....
.....[2]

(b) Explain how DNA homology is used to predict phylogeny.

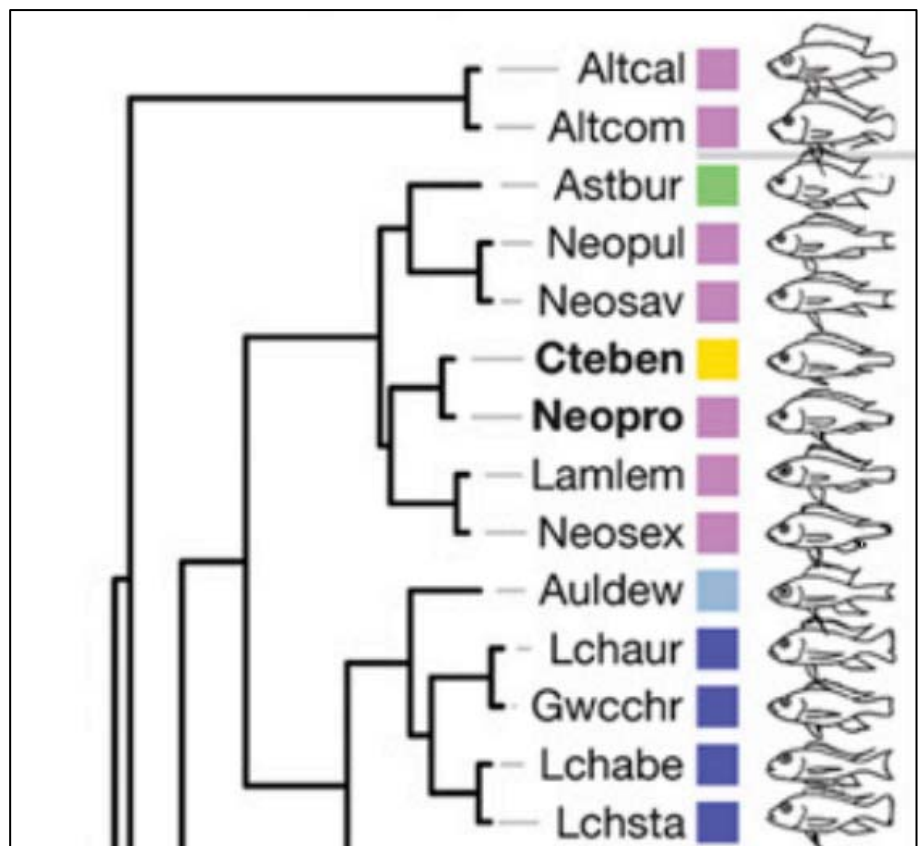
.....
.....
.....[2]

(A) Body Plan / Shape

Fig 4.3

(A) Portion of the Cluster analysis * proposing phylogenetic relationships on the basis of 17 landmarks on body shape. Outlines in (A) are based on real photographs

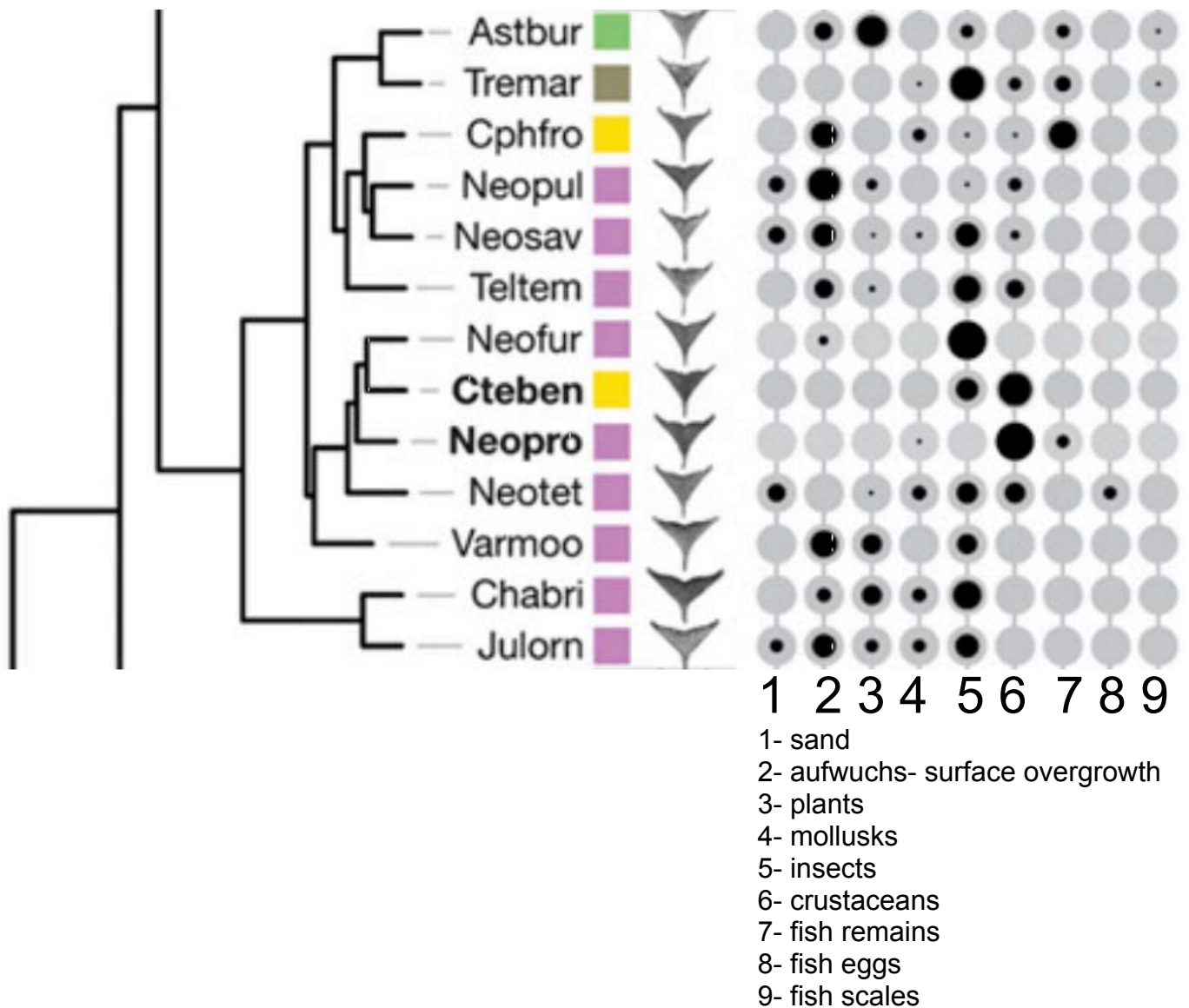
*Cluster Analysis is a systematic comparison of parameters that allows a conclusion on relationships between specimens.



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(B) lower pharyngeal jaw bone

(C) gut content



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Fig 4.4

(B) Portion of the Cluster analysis proposing phylogenetic relationships on the basis of landmarks on the lower pharyngeal jaw bone. images in (B) are taken from dissected LPJs

(C) Results from the stomach and gut content analyses (in volume %).

(c) Explain the significance of the results found in Fig 4.2, 4.3 and 4.4 on the phylogeny of 'Neopro' and 'Cteben' species.

.....

.....

.....[2]

[Total: 6]

Section B

Answer **ONE** question. Answer each part on a **separate** piece of paper.

Write your answers on separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** & **(c)** as indicated in the question.

- 5(a)** Describe how carbohydrates are formed in the Calvin cycle. [7]
- (b)** Outline the role of NAD and NADP in cells. [7]
- (c)** Compare and contrast the process of electron transport through the electron transport chain in the mitochondrion and chloroplast. [6]

OR

- 6(a)** Describe the main characteristics as well as the different developmental potentials of stem cells. Cite examples wherever appropriate. [7]
- (b)** Describe how bacteria can be used in genetic engineering to improve the nutritional content of crops. [7]
- (c)** Discuss the negative consequences of genetically-modifying organisms. [6]

[Total: 20]

END OF PAPER